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HALF YEARLY EXAMINATION - 2025

12 STD. BIOLOGY

TIME : 3 Hrs MARKS : 70

Bio Botany Marks : 35

I Answer All the questions. **8x1=8**

1. A Plant called x possesses small flower with reduced perianth and versatile anther. The probable agent for pollination would be
   a) Water b) Wind c) butterflies d) beetles
2. "Gametes are never hybrid". This is a statement of
   a) Law of dominance b) Law of independent assortment c) Law of segregation
   d) Law of random fertilization.
3. In which techniques Ethidium Bromide is used.?
   a) Southern Blotting techniques b)Western Blotting techniques
   c) Polymerase Chain Reaction d) Agarose Gel Electroporosis
4. Virus free plants are developed from
   a) Organ culture b) Meristem culture c) Protoplast culture d) Cell suspension culture
5. In soil water available for plants is
   a) gravitational water b) chemically bound water c) capillary water d) hygroscopic water
6. Which of the following is not a sedimentary cycle
   a) Nitrogen cycle b) Phosphorous cycle c) Sulphur cycle d) Calcium cycle
7. Depletion of which gas in the atmosphere can lead to an increased incidence of skin cancer.
   a) Ammonia b) Methane c) Nitrous oxide d) Ozone
8. The active principle trans-tetra hydro canabial is present in
   a) Opium b) Curcuma c)Marijuana d) Andrographis

II Answer any FOUR questions. **4x2=8**

9. Write short note on Heterostyly.
10. What is back cross?
11. What is Crossing over?
12. Name the chemicals used in gene transfer?
13. What is vivipary? Name the plant group which exhibits vivipary.
14. Enumerate the uses of turmeric.

III Answer any THREE questions. Qn. no 19 is Compulsory. **3x3=9**

15. Write the effect of Global warming in plants.
16. How is rhytidome act as the structural defence by plants against fire.
17. Write the benefits and risk of Genetically Modified foods.
18. What is Gene mapping? Write its uses.
19. Draw and label the parts of monocot seed.

IV Answer the all questions. **2x5=10**

20. a) What is endosperm.? Explain their types. ***(OR)***
    b) Describe dominant epistasis with an example.
21. a) What are the different types of hybridization? Explain. ***(OR)***
    b) How will you prepare an organic Pesticide for your home garden with the vegetables available from your kitchen?

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BIO - ZOOLOGY

Section - 1

Note : i) Answer all the questions. ii) Choose the most appropriate answer from the given four alternatives and write the option code and the corresponding answer. 8 X 1 = 8

1. Who proposed the germplasm theory?
   a) Darwin b) August Welsmann c) Lamarck d) Alfred Wallace
2. Common cold is caused by ..... .
   a) Alpha virus b) Flavi virus c) Rhino virus d) Polio virus
3. The androgen binding protein (ABP) is produced by
   a) Leydig cells b) Sertoli cells c) Hypothalamus d) Pituitary gland
4. X – chromosome was discovered by ..... .
   a) Stevens b) Henking c) Bertram d) Mary Lyon
5. Transgenic animals are those which have
   a) Foreign DNA in some of their cells b) Foreign RNA in some of their cells
   c) Foreign DNA in all their cells d) Foreign RNA in all their cells
6. The thickness of stratospheric ozone layer is measured in / on ..... .
   a) Melson units b) Sieverts unit c) Dobson units d) Beaufort scale
7. All populations in a given physical area are defined as ..... .
   a) Biome b) Ecosystem c) Territory d) Biotic factors
8. A mRNA molecule is produced by
   a) Transcription b) Replication c) Duplication d) Translation

Section - 2

Note : Answer any four of the following questions. 4 X 2 = 8

9. What are the factors that five habitat loss?
10. What is Apolysis?
11. Differentiate - leading strand and lagging strand.
12. Draw a neat labelled structure of the human ovum.
13. What is meant by sameer?
14. Expand the following : a) ZIFT b) ICSI

Section - 3

Note : Answer any three of the following questions. Question No. 19 is compulsory. 3 X 3 = 9

15. What is haplodiploidy?
16. Name the three layers of the uterine wall.
17. Define : a) Tubectomy b) Vasectomy
18. What is anaphymaxis?
19. Differentiate 'J' and 'S' shaped curves.

Section - 4

Note : Answer the following questions. 2 X 5 = 10

20. a) Differentiate between active immunity and passive immunity. ***(OR)***
    b) Explain the different kinds of syngamy in living organisms.
21. a) Explain the structure of immunoglobin with a suitable diagram. ***(OR)***
    b) List out the various causes for biodiversity losses.

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